Operation Management

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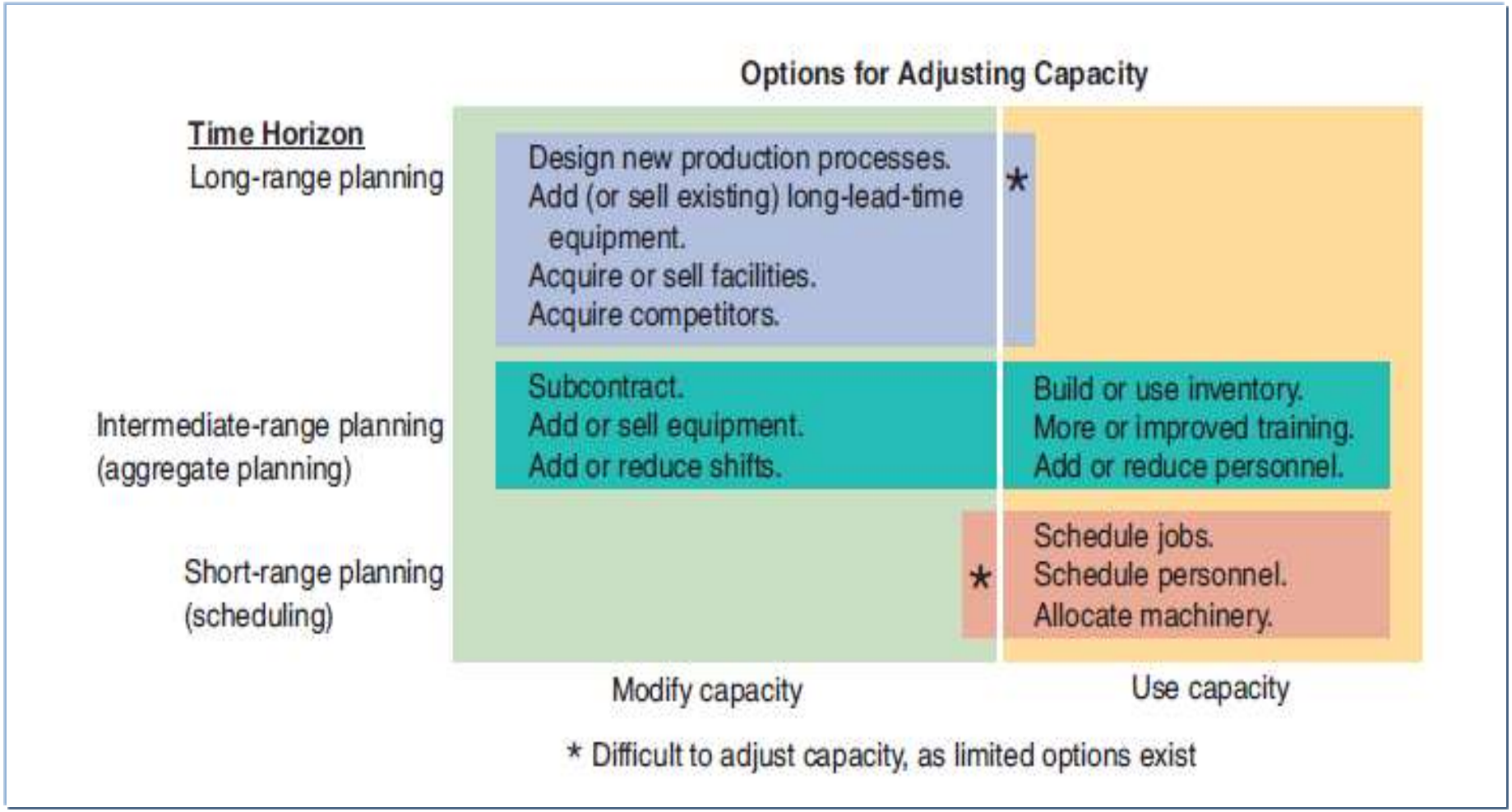
<MBACCZG526>Operation Management>
Lecture No. 5

Capacity

- Capacity is the "throughput," or the number of units a facility can hold, receive, store, or produce in a given time.
- Capacity decisions = f (Capital or Fixed Cost)
- Capacity >> Demand → Underutilized Capacity
- Capacity << Demand → Loss of Market Share.

Capacity Planning



Source: Heizer, Render, Muckson

Capacity Measurement

TABLE S7.1 Capacity Measurements

MEASURE	DEFINITION	EXAMPLE
Design capacity Andrey Eremin/123rf	Ideal conditions exist during the time that the system is available.	If machines at Frito-Lay are designed to produce 1,000 bags of chips/hr., and the plant operates 16 hrs./day. **Design Capacity = 1,000 bags/hr. × 16 hrs.** **= 16,000 bags/day**
Effective capacity Hongqi Zhang/123rf	Design capacity minus lost output because of *planned* resource unavailability (e.g., preventive maintenance, machine setups/ changeovers, changes in product mix, scheduled breaks)	If Frito-Lay loses 3 hours of output per day (namely 0.5 hrs./day on preventive maintenance + 1 hr./day on employee breaks + 1.5 hrs./day setting up machines for different products). **Effective Capacity = 16,000 bags/day − (1,000 bags/hr.)(3 hrs./day)** **= 16,000 bags/day − 3,000 bags/day** **= 13,000 bags/day**
Actual output Hongqi Zhang/123rf	Effective capacity minus lost output during *unplanned* resource idleness (e.g., absenteeism, machine breakdowns, unavailable parts, quality problems)	On average, if machines at Frito-Lay are not running 1 hr./day due to late parts and machine breakdowns. **Actual Output = 13,000 bags/day − (1,000 bags/hr.)(1 hr./day)** **= 13,000 bags/day − 1,000 bags/day** **= 12,000 bags/day**

Source: Heizer, Render, Muckson

Utilization and Efficiency

- Utilization: Actual output as a percent of design capacity.
- Efficiency : Actual output as a percent of effective capacity.

$$\text{Utilization} = \text{Actual output}/\text{Design capacity}$$

$$\text{Efficiency} = \text{Actual output}/\text{Effective capacity}$$

Class Exercise

- Sara James Bakery has a plant for processing *Deluxe* breakfast rolls and wants to better understand its capability. Last week the facility produced 148,000 rolls.
- The effective capacity is 175,000 rolls. The production line operates 7 days per week, with three 8-hour shifts per day. The line was designed to process the nut-filled, cinnamon-flavored *Deluxe* roll at a rate of 1,200 per hour.
- Determine the design capacity, utilization, and efficiency for this plant when producing this *Deluxe* roll.

Class Exercise (Expansion)

- The manager of Sara James Bakery (previous exercise) now needs to increase production of the increasingly popular *Deluxe* roll. To meet this demand, she will be adding a second production line.
- The second line has the same design capacity (201,600) and effective capacity (175,000) as the first line; however, new workers will be operating the second line.
- Quality problems and other inefficiencies stemming from the inexperienced workers are expected to reduce output on the second line to 130,000 (compared to148,000 on the first).
- The utilization and efficiency were 73.4% and 84.6%, respectively, on the first line.
- Determine the new utilization and efficiency for the *Deluxe* roll operation after adding the second line.

Any Question

Demand

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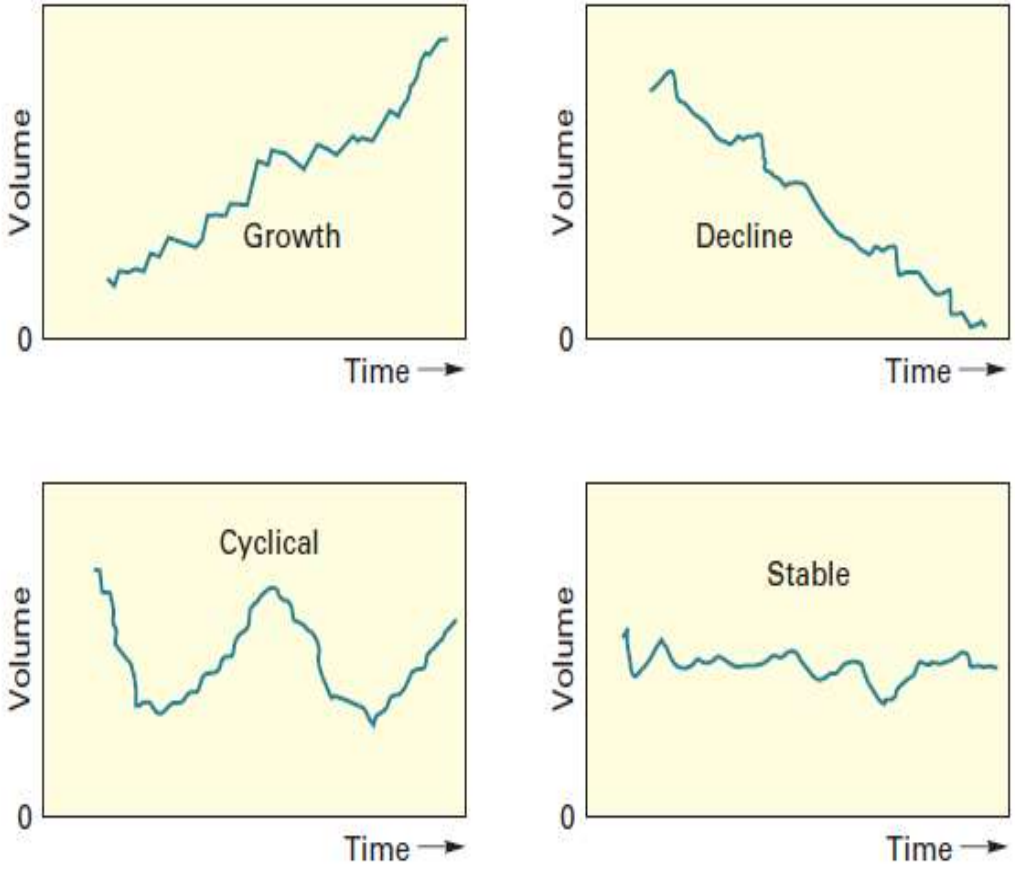


FIGURE 5.1
Common demand patterns

Source: Operation Management by William J Stevenson

Managing Demand

Demand > Capacity :

- Increased Bargaining Power of Suppliers.
- Price Increase.
- Long Lead times.
- Discourage Marginally Profitable business.
- Long term solution is increasing Capacity.

Managing Demand

Demand < Capacity :

- Increased Bargaining Power of Buyers.
- Price Decrease and Aggressive marketing.
- Accommodate the market through product changes.
- Long term solution is increasing Flexibility.

Managing Demand

Airline Industry & Seasonal Demand

OM in Action — Matching Airline Capacity to Demand

Airlines constantly struggle to control their capital expenditures and to adapt to unstable demand patterns.

Southwest and Lufthansa have each taken their own approach to increasing capacity while holding down capital investment. To manage capacity constraints on the cheap, Southwest squeezes seven flight segments out of its typical plane schedule per day—one more than most competitors. Its operations personnel find that quick ground turnaround, long a Southwest strength, is a key to this capital-saving technique.

Lufthansa has cut hundreds of millions of dollars in new jet purchases by squashing rows of seats 2 inches closer together. On the A320, for example, Lufthansa added two rows of seats, giving the plane 174 seats instead of 162. For its European fleet, this is the equivalent of having 12 more Airbus A320 jets. But Lufthansa will tell you that squeezing in more seats is not quite as bad as it sounds, as the new generation of ultra-thin seats provides passengers with more leg room. Using a strong mesh, similar to that in fancy office chairs (instead of inches of foam padding), and moving magazine pockets to the top of seat backs, there is actually more knee room than with the old chairs.

Unstable demands in the airline industry provide another capacity challenge. Seasonal patterns (e.g., fewer people fly in the winter), compounded by spikes in demand during major holidays and summer vacations, play havoc with efficient use of capacity. Airlines attack costly seasonality in several ways. First, they schedule more planes for maintenance and renovations during slow winter months, curtailing winter capacity; second, they seek out contra-seasonal routes. And when capacity is substantially above demand, placing planes in storage (as shown in the photo) may be the most economical answer.

Airlines also use revenue management (see Chapter 13) to maximize per-seat pricing of available capacity, regardless of current demand patterns.

Sources: The Wall Street Journal (February 29, 2012) and (October 6, 2011).

Source: Heizer, Render, Muckson

Matching Capacity & Demand

- Making staffing changes.
- Adjusting equipment
- Improving processes to increase throughput.
- Adding process flexibility to better meet changing product preferences.
- Closing facilities.

Class Exercise

- A department works one 8-hour shift, 250 days a year, and has these figures for usage of a machine that is currently being considered:

Product	Annual Demand	Processing Time Per Unit (hr)	Processing Time
1	400	5	
2	300	8	
3	700	2	

How many machines are required to match Capacity & Demand

Service Capacity and Demand

- In the service sector, scheduling customers is *demand management* , and scheduling the workforce is *capacity management*.
- When Demand is matching with supply, we focus on demand management.
- When Demand is not matching with supply, we focus on capacity management.

Demand Management

- Demand matches with Supply.
- Handled with appointments, reservations, and first come-first serve rule.
- Doctors, Lawyers ,etc.----Appointments.
- Rental agencies, Hotels, etc.----Reservations.
- Retail Shop, Post Offices, Restaurants----First come first serve.

Capacity Management

- Managing Demand is not feasible.
- Full time, part-time and temporary staffing.

Service Capacity Challenges

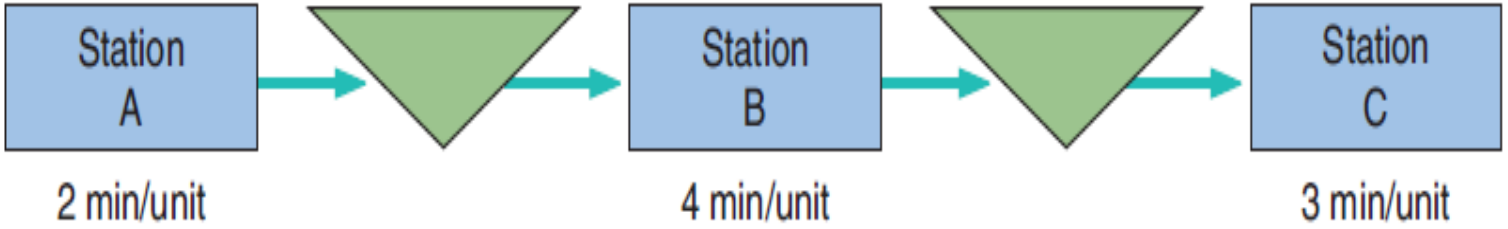
- Need to be near customers.
- Inability to store services.
- The degree of volatility of demand.

Any Question

Bottleneck Analysis

- Capacity analysis.
- Bottleneck.
- Process time.
- Bottleneck time.
- Throughput time.

Class Exercise



- What is the Process time of Process A, Process B and Process C?
- Which of the Process is a Bottleneck ?
- What is the Throughput time?

Source: Heizer, Render, Muckson

Class Exercise (Parallel Process)

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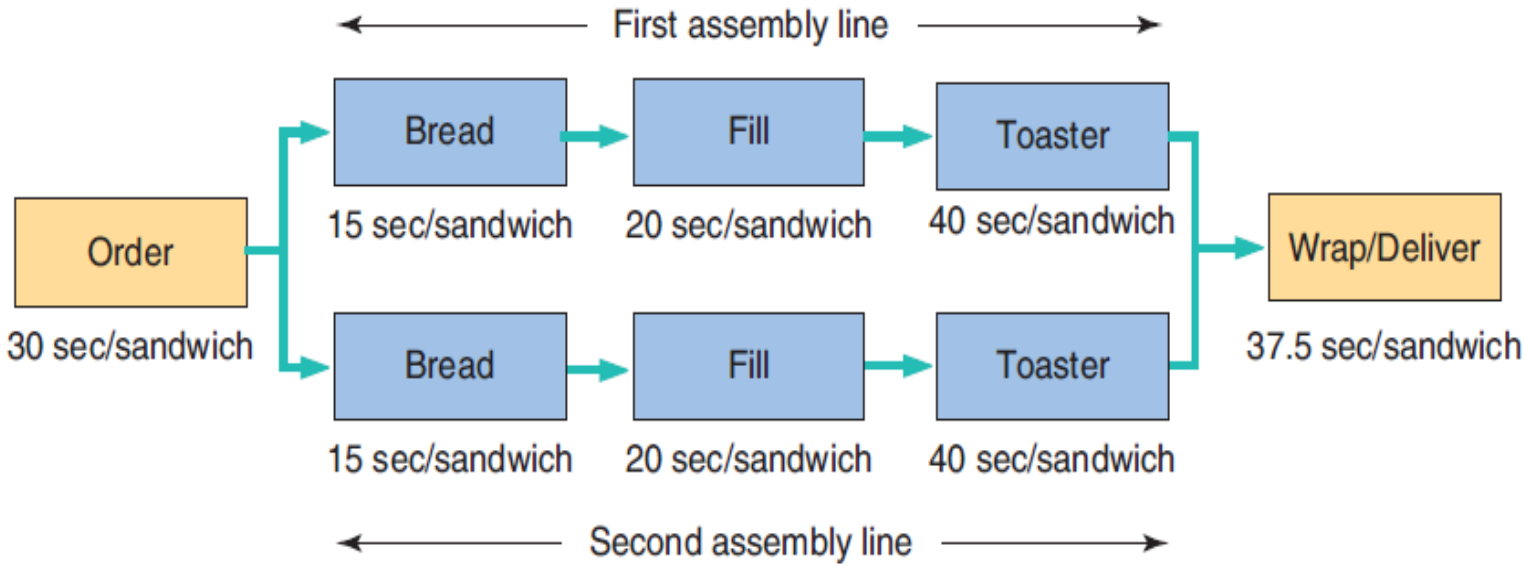
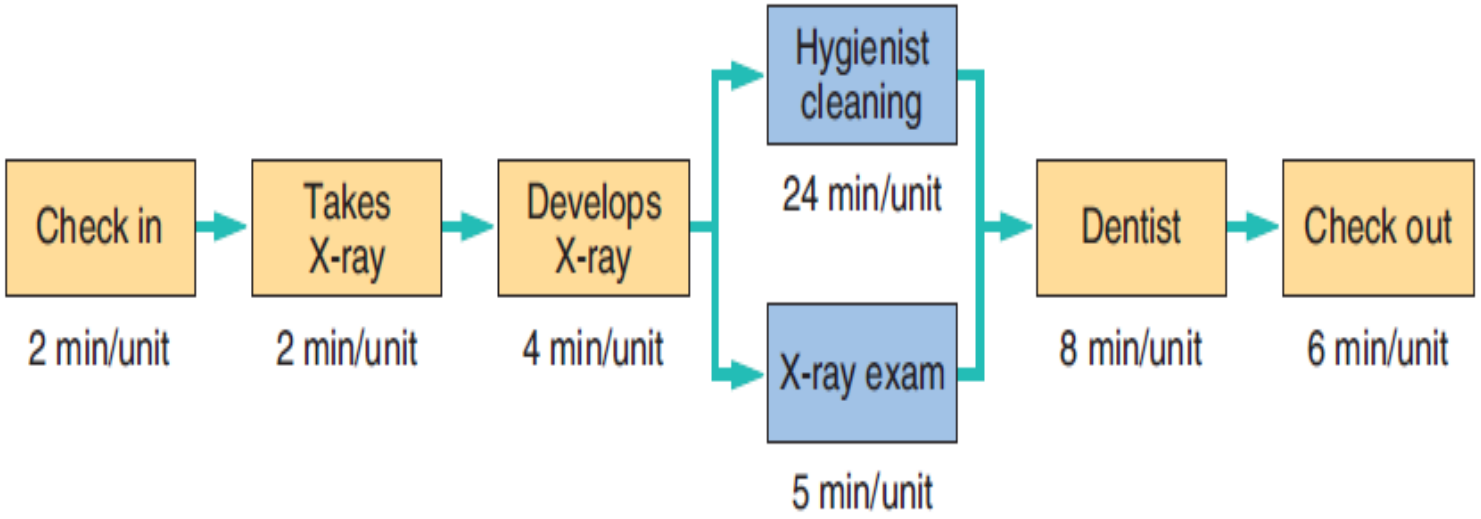


- Which of the Process is a Bottleneck ?
- What is the Throughput time?

Source: Heizer, Render, Muckson

Class Exercise (Simultaneous Process)



- Which of the Process is a Bottleneck ?
- What is the Throughput time?

Bottleneck Analysis

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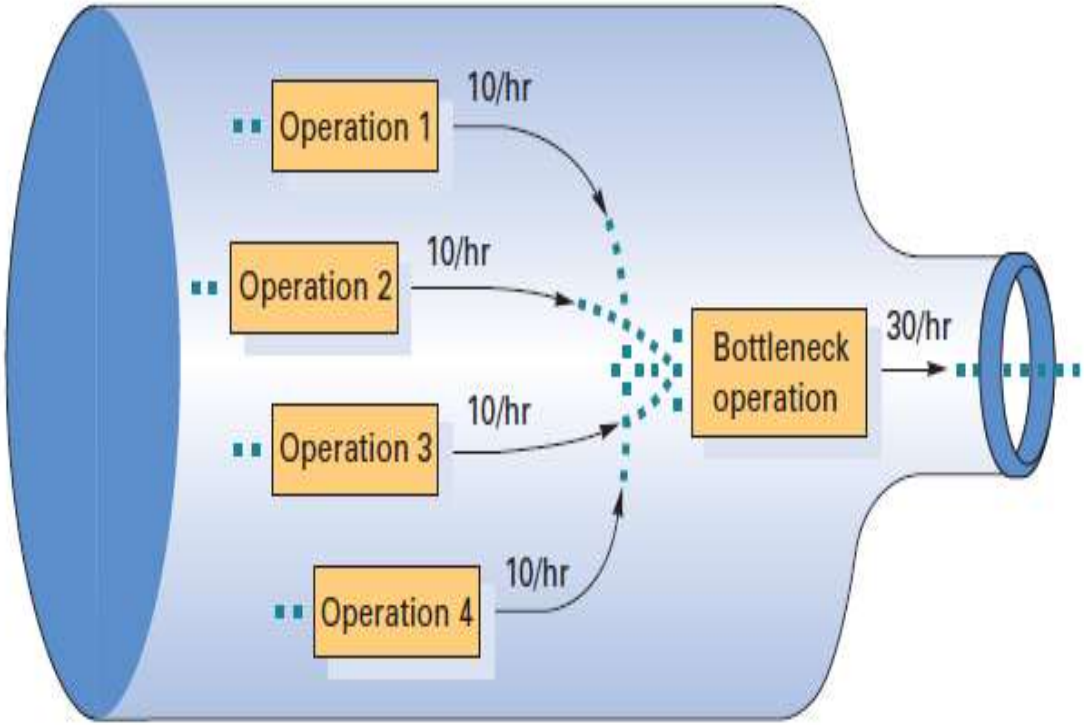

FIGURE 5.2
Bottleneck operation

Source: Operation Management by William J Stevenson

Take Home Exercise

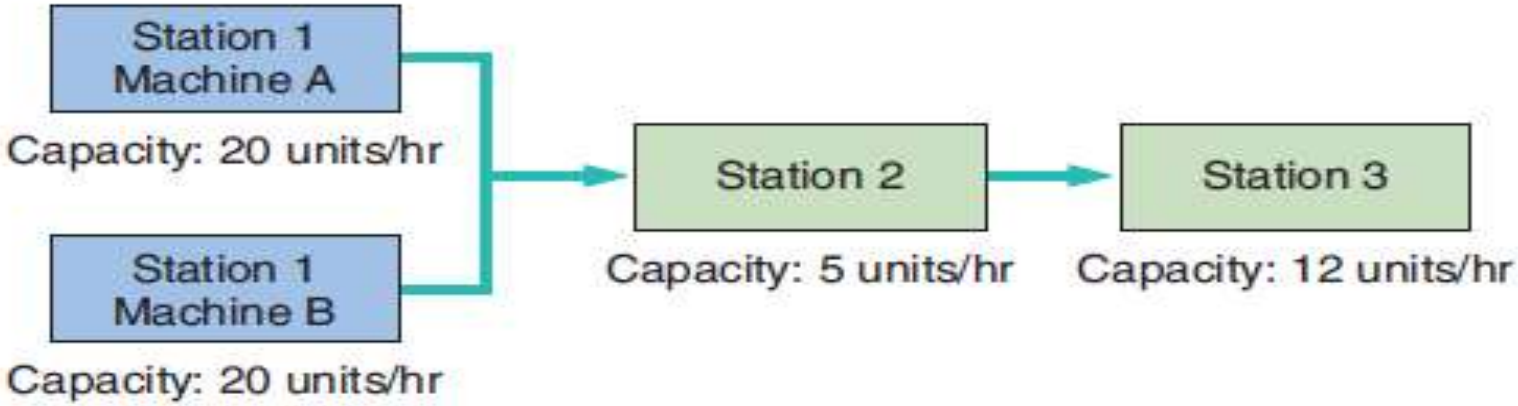

Figure S7.7

a) What is the bottleneck time of the system?
b) What is the bottleneck station of this work cell?
c) What is the throughput time?
d) If the firm operates 10 hours per day, 5 days per week, what is the weekly capacity of this work cell?

Source: Heizer, Render, Muckson

Any Question

Break Even Analysis

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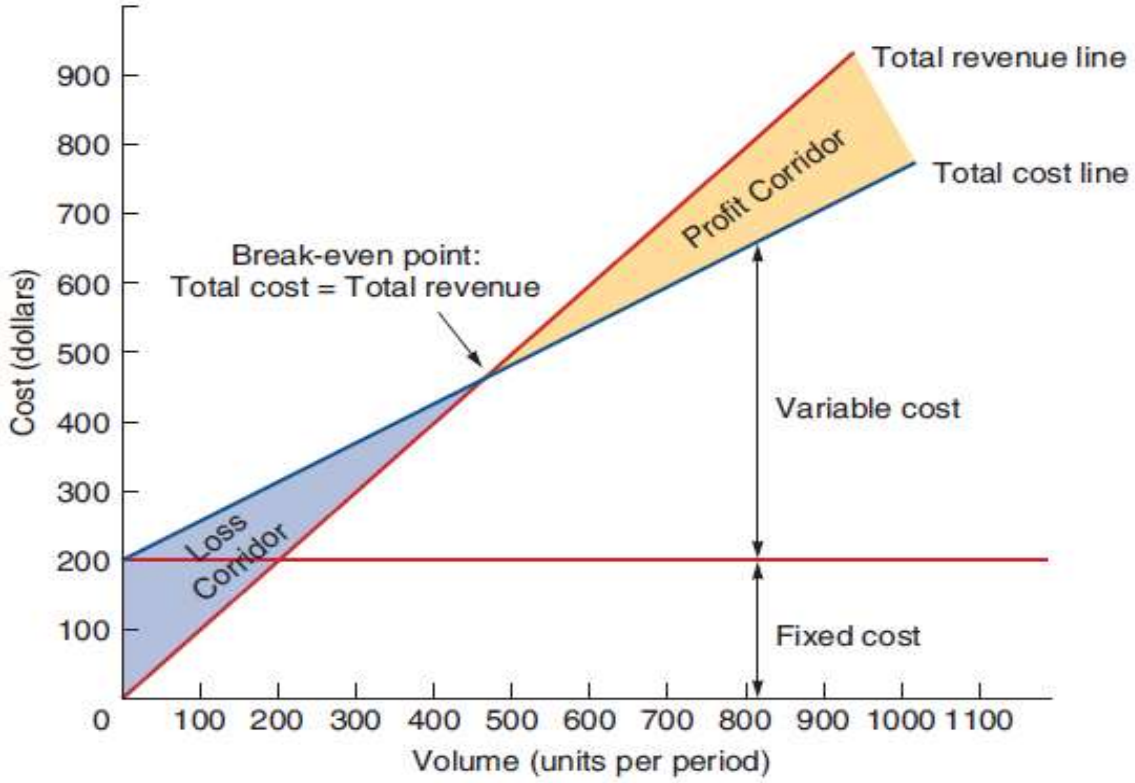

Source: Heizer, Render, Muckson

Break Even Analysis

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$$\text{Break-even in units } (BEP_x) = \frac{\text{Total fixed cost}}{\text{Price} - \text{Variable cost}} = \frac{F}{P - V}$$

$$\text{Break-even in dollars } (BEP_\$) = \frac{\text{Total fixed cost}}{1 - \dfrac{\text{Variable cost}}{\text{Price}}} = \frac{F}{1 - \dfrac{V}{P}}$$

$$\text{Break-even point in dollars } (BEP_\$) = \frac{F}{\sum\left[1 - \left(\dfrac{V_i}{P_i}\right) \times (W_i)\right]}$$

Class Exercise

- The firm first determines that it has fixed costs of $10,000 this period. Direct labor is $1.50 per unit, and material is $.75 per unit. The selling price is $4.00 per unit.

Class Exercise

MULTIPRODUCT BREAK-EVEN ANALYSIS

Le Bistro, like most other resturants, makes more than one product and would like to know its break-even point in dollars. Information for Le Bistro follows. Fixed costs are $3,000 per month.

ITEM	ANNUAL FORECASTED SALES UNITS	PRICE	COST
Sandwich	9,000	$5.00	$3.00
Drinks	9,000	1.50	0.50
Baked potato	7,000	2.00	1.00

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Item	Annual Forecasted Sales Unit	Price	Cost (V)	V/P	1-V/P	Annual Forecasted Sales	% of Sales	Weigted Contribution
Sandwich	9000	5	3	0.6	0.4	45000	0.62068966	0.248275862
Drinks	9000	1.5	0.5	0.333333	0.666667	13500	0.1862069	0.124137931
Baked Potato	7000	2	1	0.5	0.5	14000	0.19310345	0.096551724
						72500	1	0.468965517
						Break Even	76764.7059	

Any Question

Theory of Constraint (TOC)

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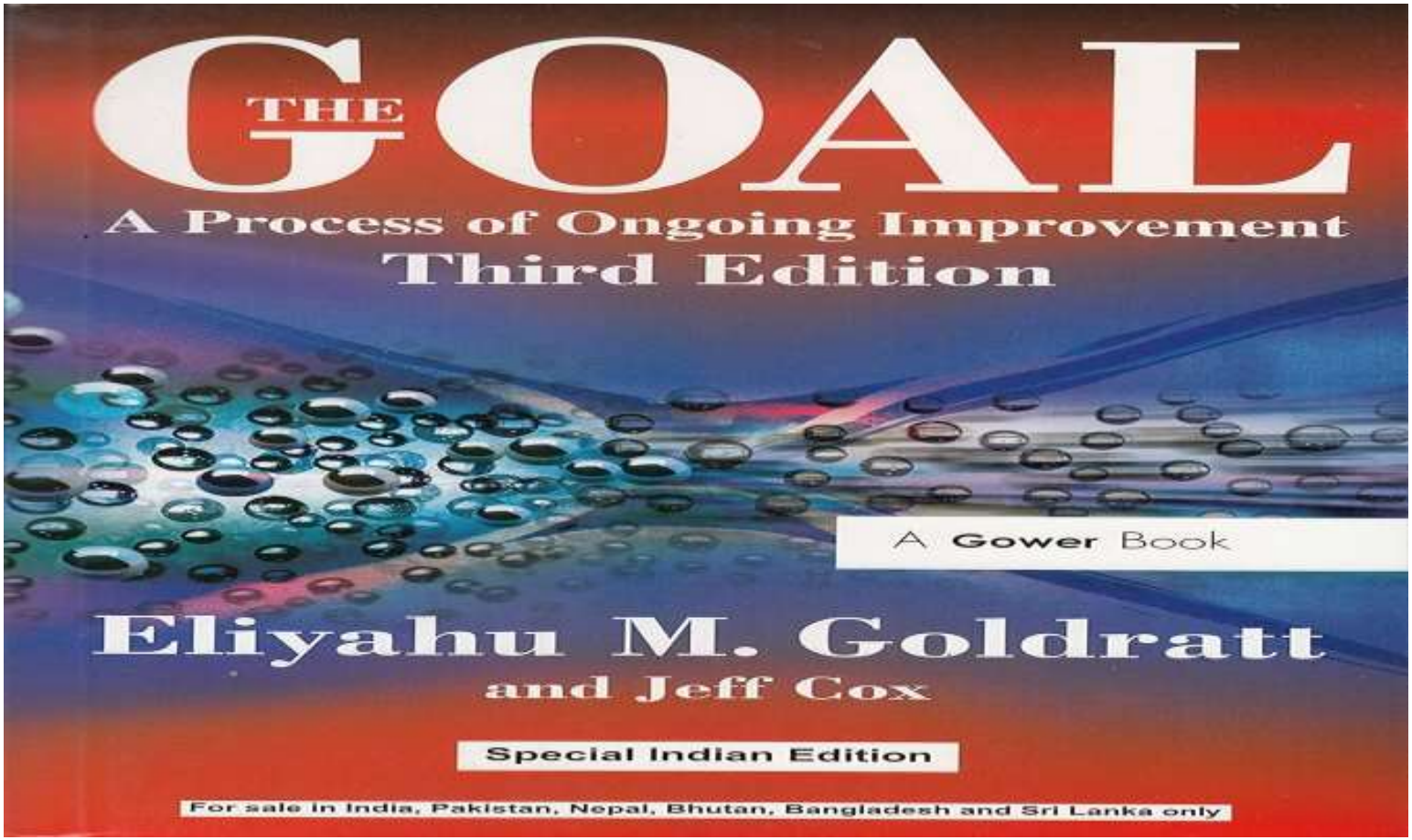

Source: Amazon.in

- The goal of every business organization is to make money, both now and in future.
- Throughput: Quantity of money that the organization generates through sales.
- Inventory: Quantity of money invested in materials that the organization intends to sell.
- Operating expense: Quantity of money spent in converting inventory to throughput.

Types of Constraint

- Market: Insufficient demand.
- Resource: Too little of one or more resources (e.g., workers, equipment, and space).
- Material: Too little of one or more materials.
- Financial: Insufficient funds.
- Supplier: Unreliable, long lead time, substandard quality.
- Knowledge or competency: Needed knowledge or skills missing or incomplete.
- Policy: Laws or regulations interfere.

Class Exercise

A company makes a single product whose weekly demand is 100. Identify the constraint in each of the mentioned situations :

- The company works 8-hour shifts (5 days) and takes 20 minutes to make the product.
- The company takes 30 minutes to make each product.
- The supplier can give raw material at the rate of 15/day.
- The supplier can give 25 units per day, but the organization takes two days to place the order.

Bottleneck Management

1. Release work orders to the system at the pace set by the bottleneck's capacity.

- The theory of constraints utilizes the concept of drum, buffer, rope.
- Drum: It is the beat of the system. It provides the schedule—the pace of production.

- Buffer : It is the resource, usually inventory, which may be helpful to keep the bottleneck operating at the pace of the drum.
- Rope: Rope provides the synchronization or communication necessary to pull units through the system. The rope can be thought of as signals between workstations.

2. Lost time at the bottleneck represents lost capacity for the whole system.
3. Increasing the capacity of a non-bottleneck station is a mirage.
4. Increasing the capacity of the bottleneck increases capacity for the whole system.

Reducing Risk with Incremental Changes

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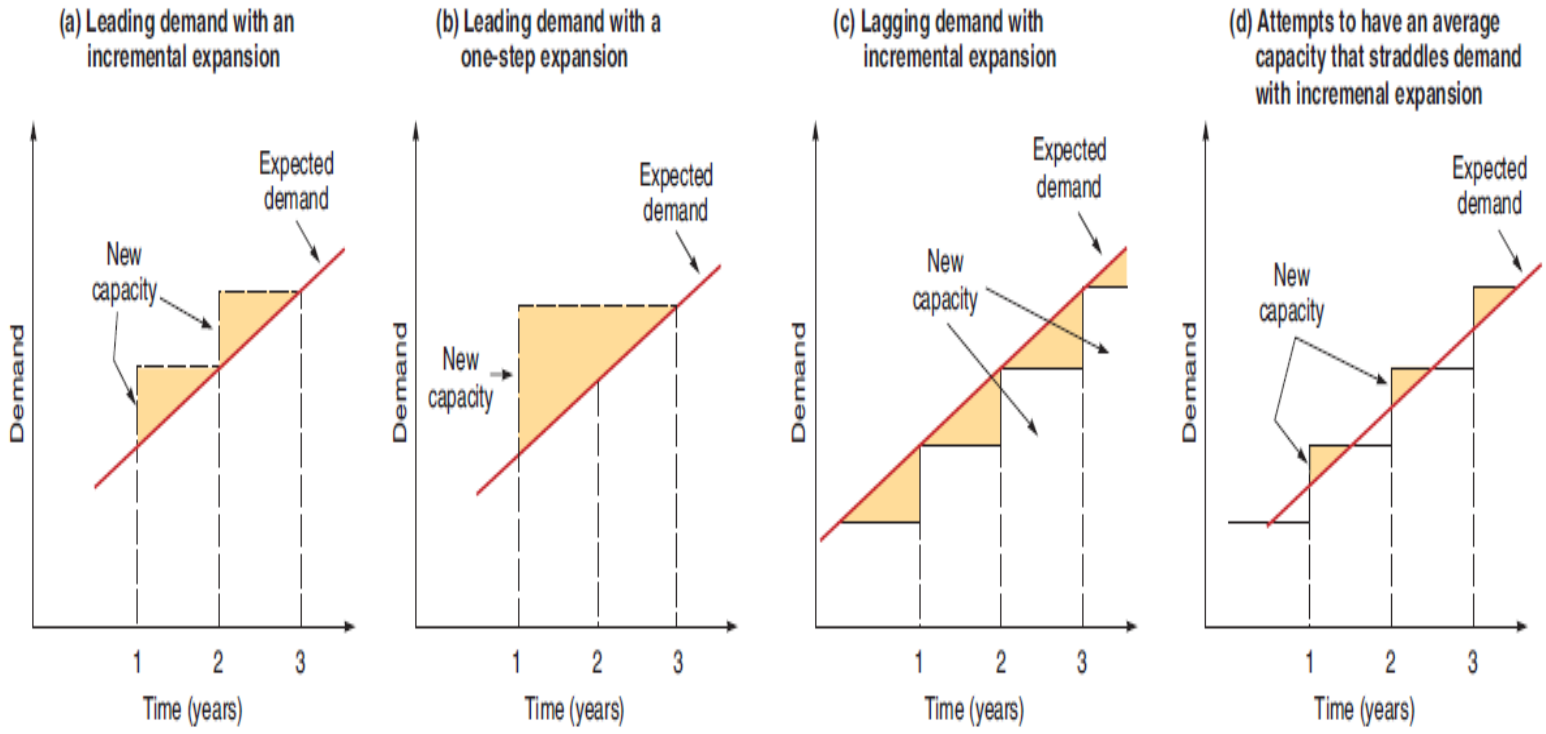

Source: Heizer, Render, Muckson

Expected Monetary Value (EMV)

- Capacity Expansion.
- Determining expected monetary value (EMV) requires specifying alternatives and various states of nature.
- By assigning probability values to the various states of nature, we can make decisions.

Class Exercise

- Southern Hospital Supplies, a company that makes hospital gowns, is considering capacity expansion.
- Southern's major alternatives are to do nothing, build a small plant, build a medium plant, or build a large plant. The new facility would produce a new type of gown, and currently the potential or marketability for this product is unknown.
- If a large plant is built and a favorable market exists, a profit of $100,000 could be realized. An unfavourable market would yield a $90,000 loss. However, a medium plant would earn a $60,000 profit with a favourable market. A $10,000 loss would result from an unfavourable market. A small plant, on the other hand, would return $40,000 with favourable market conditions and lose only $5,000 in an unfavourable market. Of course, there is always the option of doing nothing.

- Recent market research indicates that there is a .4 probability of a favourable market, which means that there is also a .6 probability of an unfavourable market.
- With this information, the alternative that will result in the highest expected monetary value (EMV) can be selected.

Solution

Compute the EMV for each alternative:

EMV (large plant) = (.4) (100,000) + (.6) (-90,000) = - 14,000

EMV (medium plant) = (.4) (60,000) + (.6) (-10,000) = +18,000

EMV (small plant) = (.4) (40,000) + (.6) (- 5,000) = + 13,000

EMV (do nothing) = +0

Based on EMV criteria, Southern should build a medium plant.

Any Question

Further Reading / References

1. **Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 7**
2. Operation Management by William J Stevenson—Chapter 5

THANK YOU